Chapter 3: Important Insect Pests of Crops & Storage

Introduction

- **Entomology:** The branch of zoology that deals with the study of insects.
- Insects belong to the Phylum **Arthropoda** and Class **Insecta**.
- The insect's body is divided into three parts: **Head, Thorax, and Abdomen**.
- Their legs and body are multi-segmented. The body is actually made up of 20 segments (the first 6 form the head, the next 3 form the thorax, and the last 11 form the abdomen).
- **Main Morphological Characteristics:**
    - **Head:** Possesses one pair of antennae, one pair of compound eyes, and mouthparts.
    - **Thorax:** Has three pairs of jointed legs (Hexapoda) and mostly two pairs of wings (sometimes absent or one pair), usually located on the 2nd and 3rd thoracic segments.
    - **Respiration & Excretion:** Respiration occurs through branched tracheae, and excretion through Malpighian tubules.
    - **Body Cavity:** It acts as a blood cavity (Haemocoel).
    - **Skeleton:** They have no bones; their exoskeleton is made of a chitinous covering.
    - **Reproduction:** They are unisexual (male and female are separate) and mostly oviparous (egg-laying).
- **Metamorphosis:** Occurs after embryonic development.
    - **Complete Metamorphosis:** Egg → Larva → Pupa → Adult (4 stages).
    - **Incomplete Metamorphosis:** Egg → Nymph → Adult (3 stages).

Classification of Crop Insects

1. Based on Seasons:

- **(i) Kharif Crop Pests:** Red hairy caterpillar (Katra), Kharif grasshopper, White grub, Armyworm, etc.
- **(ii) Rabi Crop Pests:** Wheat pink borer, Mustard aphid, Gram pod borer, etc.

2. Based on Crops:

- **(i) Cereal Pests:** Katra, White grub, Termite, Shoot fly, Stem borer.
- **(ii) Pulse Pests:** Gram pod borer, Cutworm, Leaf miner, Plume moth.
- **(iii) Oilseed Pests:** Mustard aphid, Sesame capsule borer, Groundnut leaf miner.
- **(iv) Vegetable Pests:** Pea leaf miner, Diamondback moth, Flea beetle, Whitefly, Brinjal stem and fruit borer.
- **(v) Fruit Pests:** Mango hopper, Bark eating caterpillar, Lemon butterfly.
- **(vi) Storage Pests:** Khapra beetle, Pulse beetle, Rice weevil, Rust-red flour beetle.

3. Based on Economics:

- **(i) Beneficial Insects:** Honey bee, Silkworm, Ladybird beetle, *Chrysoperla carnea*, Lac insect.
- **(ii) Harmful Insects:** Pod borer, Diamondback moth, Termite, Grasshopper.

4. Based on Science (Orders): Divided into 29 orders. Important agricultural orders:

1. Odonata (Dragonfly)
2. Plecoptera (Stonefly)
3. Orthoptera (Grasshopper/Locust)
4. Dictyoptera (Cockroach, Mantids)
5. Isoptera (Termite)
6. Mallophaga (Bird lice)
7. Hemiptera (Bugs)
8. Thysanoptera (Thrips)

9. Mecoptera (Scorpionfly)
10. Lepidoptera (Butterfly, Moth)
11. Diptera (Flies, Mosquitoes)
12. Hymenoptera (Honey bees, Wasps)
13. Coleoptera (Beetles, Weevils)

1. Red Hairy Caterpillar (Katra)

- **Scientific Name:** *Amsacta moorei*
- **Order:** Lepidoptera | **Family:** Arctiidae
- **Nature:** It is a polyphagous insect found abundantly in Northern India.

Characteristics / Symptoms:

- The adult moth is stout, medium-sized, with white wings featuring black spots.
- The outer margins of the forewings, a part of the thorax, and the abdomen are scarlet-red in color.
- The fully developed caterpillar (larva) is 40 to 50 mm long. Its entire body is covered with reddish-brown hairs, and its head is dark brown.

Host Plants:

- It causes maximum damage to almost all Kharif crops, especially pearl millet (bajra), maize, sorghum, cluster bean (guar), moong, moth bean, sunhemp, and sesame.

Damage and Importance:

- The outbreak is severe in light soils and low rainfall regions.
- The caterpillars feed in groups on tender plant parts. Upon growing, they separate and devour leaves and soft parts, completely destroying the plant. Severe outbreaks can cause 100% crop loss.

Life Cycle:

- Adult moths emerge from the soil at night after the first one or two heavy showers of the monsoon and are attracted to light sources.
- A few hours after mating, the female moth lays pale yellow, opium seed-like eggs in clusters on the lower surface of leaves. One female can lay up to 1500 eggs.
- Larvae emerge in 4-5 days. After molting 6 times, they fully develop in 15-23 days.
- Before pupating, they move to bunds, bushes, or under trees and turn into pupae at a depth of 10-30 cm in the soil.
- They remain dormant as pupae for 9-10 months until the next monsoon. Total life cycle is 26-41 days (only 1 generation per year).

Management:

- **Cultural:** Set up light traps after the first monsoon rain to attract and kill adult moths. Destroy egg clusters before hatching.
- **Biological:** Deep summer ploughing exposes pupae to predatory birds (myna, crow, egret) and high temperatures. Egg parasitoids like *Trichogramma* and *Telenomus* can be used.
- **Chemical:** Dust Malathion 5% or Quinalphos 1.5% @ 25 kg/ha, or spray Quinalphos 25 EC (1.25 liters in 250-500L water/ha). Dig a 30x30 cm trench around the field and treat with Quinalphos dust to trap and kill migrating caterpillars.

2. White Grub

- **Scientific Name:** *Holotrichia consanguinea*
- **Order:** Coleoptera | **Family:** Melolonthidae

- **Nature:** A pest of national importance. Also known as Cockchafer Beetles, May Beetles, or June Beetles.

Characteristics:

- Adult beetle is dark brown, 18 mm long and 7 mm wide. It has two pairs of wings; the thick forewings cover the hindwings during resting.
- The grub is white with a brown head and strong jaws. The infant grub is 10-12 mm long, and its body is "C-shaped".

Host Plants:

- Kharif crops (groundnut, pearl millet, chilli, guar, moong, cowpea, vegetables, fruit nurseries). Beetles feed on host trees like ber, khejri, neem, gular, and babool.

Damage and Importance:

- It is a subterranean pest. Grubs damage the roots of crops and nurseries.
- Initially feeds on decaying matter, then attacks roots. Plants turn yellow, dry up, and can be easily pulled out by hand.
- Crops with a tap root system (groundnut, chilli) are affected more than adventitious root crops (pearl millet, sorghum).
- Causes 20-100% yield loss. Maximum damage is done by the third-instar grub in August. Beetles are also polyphagous (ber is their favorite host).

Life Cycle:

- Adult beetles emerge from the soil around 7:30 PM after pre-monsoon or monsoon showers (mid-May onwards) to mate and feed on host trees.
- Females lay 50-60 eggs individually at a depth of 10-15 cm. Incubation period is 7-13 days.
- Grubs remain active from July to mid-October, then move deeper (40-70 cm) to pupate. The larval period is 82-113 days.
- Pupal period is 2-3 weeks, after which they transform into beetles and stay dormant in the soil until the next May. (1 generation per year).

Management:

- **Cultural:** Install light traps/pheromone traps and manually collect/kill beetles from host trees at night (8:30 PM - 11:30 PM) for 7-8 days after the first rain.
- **Chemical:** Seed treatment with Chlorpyriphos 20 EC or Quinalphos 25 EC. Spray host trees with Monocrotophos (1.25 ml/L). Soil application of Quinalphos 5% G or Carbofuran 3% G @ 20-25 kg/ha before sowing. In standing crops, apply Chlorpyriphos 20 EC @ 4.0 L/ha with irrigation water.

3. Locust & Grasshopper

- **Scientific Name:** *Schistocerca gregaria*
- **Order:** Orthoptera | **Family:** Acrididae
- **Nature:** Polyphagous and international pest. Out of 5000 species worldwide, only 9 are termed 'locusts'.

Three Main Species in India:

1. **Desert Locust (*Schistocerca gregaria*):** Highly destructive. Found in 70 countries.
2. **Migratory Locust (*Locusta migratoria*):** Found in Rajasthan and Gujarat. Breeds twice a year.
3. **Bombay Locust (*Patanga succincta*):** Active from Gujarat to Tamil Nadu. Smaller than the desert locust; 1 generation per year.

Phase Theory: Desert locusts exist in two main phases:

1. **Solitary Phase:** Low population. Nymphs are green, adults are grey. Behave like ordinary grasshoppers.
2. **Gregarious Phase:** Nymphs/adults in huge swarms. Nymphs are black. They travel long distances. Immature adults are pink, mature adults are yellow.*(The transitional state between solitary and gregarious is called the Transient Phase).*

Host Plants:

- Almost all Kharif crops (bajra, groundnut, chilli, vegetables, fruit nurseries).

Damage and Importance:

- Both nymphs and adults are destructive. They defoliate all vegetation completely, avoiding only specific plants like Aak, Neem, Datura, Jamun, Shisham, and Fig.

Life Cycle:

- Sexually mature adults are yellow. Females prefer sandy soil to lay eggs (10-15 cm deep) in pods containing 50-150 eggs. One female lays up to 500 eggs.
- Eggs look like yellow rice grains. Incubation period is 10-40 days.
- Vermiform larvae emerge and immediately molt into nymphs. Nymphs molt 5 times over 3-10 weeks to become adults. Adults mature sexually in 10-15 days. Breeds twice a year (summer and winter).

Management:

- **Cultural:** Collect nymphs and adults at night/early morning and destroy them using flame throwers. Spray 0.1% Neem oil on crops.
- **Biological:** *Troxprocerus* beetle eats its eggs. *Bacillus thuringiensis* bacteria is effective. Protect predatory birds like Myna and Starling.
- **Chemical:** Dig trenches around breeding sites and dust with 2% Lindane. Spread poison bait (wheat bran + sodium fluosilicate/paris green + molasses) in the morning/evening.

4. Gram Pod Borer

- **Scientific Name:** *Helicoverpa armigera*
- **Order:** Lepidoptera | **Family:** Noctuidae
- **Nature:** Polyphagous, cosmopolitan pest. Known as "Cotton bollworm" in America.

Characteristics:

- Adult moths are stout, olive-grey, or reddish-brown nocturnal fliers.
- Fully developed caterpillars are 35 mm long, light pale-green or pale-brown, with longitudinal bands on the sides.

Host Plants:

- Gram (chickpea), pea, cotton, pigeon pea, raw maize cobs, sorghum, sunflower, okra, tomato, and tobacco.

Damage and Importance:

- Active almost year-round. Initially damages leaves and tender twigs, then attacks buds, flowers, and pods.
- The larva bores into the pod (keeping half its body outside) and eats the seeds, making the pod hollow. One larva destroys 40-50 pods. Causes 30-60% fruit loss in tomatoes.

Life Cycle:

- Adult moths appear in gram crops in early November. Females lay 500-1000 eggs individually on tender plant parts.

- Larvae hatch in 2-4 days, molt 5 times, and mature in 13-20 days.
- They drop to the soil, make an earthen cell 8-12 cm deep, and pupate. The pupal stage lasts 8-15 days.

Management:

- **Cultural:** Deep summer ploughing (May-June) to destroy pupae. Use light/pheromone traps. Plant African tall marigold as a trap crop with tomatoes (1:14 ratio).
- **Chemical:** Dust Malathion 5% or Quinalphos 1.5% @ 20-25 kg/ha, or spray Quinalphos 25 EC (1.25 L/ha) at the flowering stage. Repeat after 15 days.

5. Wheat Stem Borer

- **Scientific Name:** *Sesamia inferens*
- **Order:** Lepidoptera | **Family:** Noctuidae
- **Nature:** Also known as the Pink Borer. Found mainly in UP, MP, Bihar, Rajasthan, and Tamil Nadu.

Characteristics:

- Adult moth is whitish to brown, 10-12 mm long, with an 18-20 mm wingspan. Forewings have black lines on margins, hindwings are white.

Host Plants:

- Paddy, maize, wheat, barley, sorghum, sugarcane, ragi.

Damage and Importance:

- The larval stage is destructive. It enters the stem causing the central shoot to dry up, forming a **"Dead heart"**.
- In later stages, it causes dry and empty ears (white ears). Active on sugarcane (April-July), paddy (July-Oct), and attacks wheat, maize, and sorghum during dry winter weather in North India.

Life Cycle:

- Four stages. Female lays up to 400 yellowish-white eggs in clusters inside leaf sheaths. Incubation is 7-10 days.
- Mature larva is 25 mm long, pearl-colored or purplish-pink, smooth, and cylindrical with a black head. Larval period is 20-30 days.
- Pupates inside the stem. Pupa is brown (12 mm). Pupal stage is 7-12 days. Four generations in wheat. Hibernates as larva in winter.

Management:

- **Cultural:** Adopt crop rotation. Do not sow wheat after maize or sorghum. Uproot and destroy plants with 'dead hearts'.
- **Biological:** Use natural enemies like *Bracon chilonis* and *Bracon gelichiae*.
- **Chemical:** Spray Chlorpyriphos 20 EC or Quinalphos 25 EC (0.05% solution) upon pest outbreak.

6. Fenugreek and Mustard Aphid

- **Scientific Name:** *Lipaphis erysimi*
- **Order:** Hemiptera | **Family:** Aphididae
- **Distribution:** UP, MP, Punjab, Maharashtra, Bihar, Gujarat, Karnataka, and Rajasthan.

Characteristics:

- Adults can be winged or wingless. Small size (1/10 cm), soft-bodied, green or light slate-colored.
- Abdomen has nine segments with wax glands. The last segment has the anus from which honeydew is secreted.

Damage and Importance:

- Appears from the last week of Nov to first week of Dec (peaks in Feb). Both nymphs and adults suck sap from tender stems, leaves, and pods.
- Plants turn yellow, dry up, and branching is stunted. The honeydew secretion attracts black sooty mould, further reducing photosynthesis. Reduces oil content and pod formation.

Life Cycle:

- In plains, females do not lay eggs; they give birth directly to nymphs (Ovoviviparity). In hilly areas, all 3 stages (egg, nymph, adult) are found.
- Females reproduce via mating or without mating (Parthenogenesis). Nymphs mature into adults in 3-7 days. Multiple generations in plains. Winged forms develop in February to migrate.

Management:

- **Cultural:** Early sowing of mustard (so plants can tolerate late-season attacks). Use resistant varieties (AGH-1, Type 101).
- **Biological:** Conserve natural enemies like Ladybird beetle (*Coccinella septempunctata*) and Aphid lion (*Chrysoperla carnea*).
- **Chemical:** Dust Malathion 5% or Quinalphos 5% @ 20-25 kg/ha. Alternatively, spray Dimethoate 30 EC (1 L/ha), Thiamethoxam 25 WG (100 g/ha), or Acephate 75 SP (750 g/ha).

7. Termite

- **Scientific Name:** *Odontotermes obesus*
- **Order:** Isoptera | **Family:** Termitidae
- **Nature:** Known as "White Ant". Social insect living in colonies inside a termitaria.

Characteristics (Castes):

- **Workers (80-90%):** Strong mouthparts. Responsible for foraging, building the termitaria, and feeding the colony.
- **Soldiers (3-5%):** Oversized mouthparts for colony defense.
- **King & Queen:** Royal members. Queen is the founder, lives 5-10 years, and lays 70-80 thousand eggs per day (Egg-laying machine). King fertilizes the queen.
- Reproductive castes: Macropterous, Brachypterous, and Apterous.

Damage and Importance:

- Polyphagous. Damage done strictly by nymphs and workers.
- Severe on nurseries, transplanted crops, and barren land crops. Feeds on roots and makes mud galleries on stems to eat bark from the inside.
- More damage in non-irrigated areas. Infested plants can be easily pulled out. Active year-round but subsides during heavy rains.

Life Cycle:

- Macropterous reproductives fly out after the first heavy monsoon shower (Nuptial flight).

- After mating, they build nuptial chambers in the soil. Eggs are pale yellow and kidney-shaped.
- Nymphs hatch and become workers/soldiers in 6 months. Reproductives take 1-2 years to mature. One generation per year.

Management:

- **Cultural:** Destroy termitaria and kill the queen. Do not use undecomposed green manure/cow dung. Destroy crop residues.
- **Chemical:** Treat soil with Malathion powder @ 25kg/ha before sowing. Seed treatment with Chlorpyriphos 20 EC or Quinalphos 25 EC (400 ml/100 kg seeds). Apply Chlorpyriphos 20 EC @ 4 L/ha with irrigation water for standing crops. Treat pits/stems in orchards with Malathion.

8. Khapra Beetle

- **Scientific Name:** *Trogoderma granarium*
- **Order:** Coleoptera | **Family:** Dermestidae
- **Origin:** India. Abundant globally at 32-44°C.

Characteristics:

- Mature grub is brown, covered with long stiff hairs, 4.0 mm long.
- Adult beetle is dark brown, 2-3 mm. Male is darker and half the size of the female.

Damage and Importance:

- Damages stored wheat, millet, barley, maize, paddy, pulses, and dry fruits.
- Only the grub causes damage. Maximum damage occurs from July to October (warm & moist).
- Eats the embryo, destroying germination ability and reducing nutritional value. Infestation is limited to the top 50 cm of the grain heap.

Life Cycle:

- Female lays 120-150 eggs individually or in clusters on grains.
- Incubation is 3-5 days (summer). Larval period is 20-30 days (males) to 20-40 days (females).
- Pupates in the cast skin of the last instar grub. Pupal period is 4-6 days. 4-5 generations/year.

Management:

- **Cultural:** Clean godowns. Seal crevices/holes with cement. Sun-dry harvested grains to bring moisture below 8-10%. Store gunny bags on wooden planks 50 cm away from walls. Mix neem seed powder (100:1) with grains.
- **Chemical:** Fumigate warehouses with EDCT mixture (30-40 kg/100 sq. m) or Methyl bromide (3.5 kg/100 sq. m). Do not use on milled flour.

9. Ber Fruit Fly

- **Scientific Name:** *Carpomyia vesuviana*
- **Order:** Diptera | **Family:** Tephritidae
- **Nature:** Monophagous pest of *Ziziphus* species.

Characteristics:

- Adult fly is brownish-yellow with longitudinal brown stripes on the thorax and spotted wings.

Damage and Importance:

- Only the maggot causes damage.
- Maggots feed on the pulp, causing the fruit to rot around the seed, emit a foul smell, and drop prematurely. Causes 30-100% crop loss.

Life Cycle:

- Hibernates as a pupa in the soil from April to August.
- Flies emerge during flowering/fruiting. Female bores into young fruit to lay 1-2 spindle-shaped eggs (1.0 mm).
- Maggots hatch in 2-3 days, feed for 7-10 days (up to 18 maggots per fruit).
- They exit the fruit, drop to the ground, and pupate in the soil (6-15 cm deep) for 14-30 days. Smallest cycle is 24 days. 2-3 generations/year.

Management:

- **Cultural:** Deep hoeing around trees in May-June and Dec-Jan to expose pupae. Collect and destroy/bury damaged fruits daily.
- **Chemical:** When fruits are pea-sized, spray Methyl demeton 25 EC or Dimethoate 30 EC (1 ml/L water). Repeat in mid-December.

10. Pomegranate Butterfly

- **Scientific Name:** *Deudorix (Virachola) isocrates*
- **Order:** Lepidoptera | **Family:** Lycaenidae
- **Nature:** Polyphagous (attacks apple, ber, citrus, guava, pomegranate).

Characteristics:

- Adult male is bright blue; female is brown. Wingspan 40-50 mm.
- Fully developed larva is 17-20 mm, dark brown, covered with short hairs and white patches.

Damage and Importance:

- Larvae bore into developing fruits and feed on pulp and seeds (up to 8 larvae per fruit).
- Secondary infection by bacteria/fungi causes rotting and a foul smell. Infested fruits fall off. Causes 40-90% loss.

Life Cycle:

- Breeds year-round. Female lays shiny white, oval eggs singly on the calyx or small fruits.
- Larvae hatch in 7-10 days and feed inside the fruit for 18-47 days.
- Before pupating, caterpillars tie the fruit stalk to the branch with silken strands so it doesn't fall, then pupate inside the fruit or on the stalk. Pupal stage is 7-34 days. 4 overlapping generations/year.

Management:

- **Cultural:** Cover developing fruits with bags. Collect and destroy infested fruits.
- **Chemical:** Spray Dimethoate 30 EC or Quinalphos 25 EC (2 ml/L water) to protect against attacks.